

3-17_workflow

February 27, 2021

```
[1]: import pandas as pd
import math

[2]: sheet_lst = ['sales', 'total_store_list', 'treatment_control_pairs', 'step_1']
sheet_dic = {}

for sheet in sheet_lst:
    sheet_dic[sheet] = pd.read_csv(f'materials/3-17_ab-test-walkthrough/
    ↳sales-raw-{sheet}.csv')

[3]: tmp_df = sheet_dic['step_1']
tmp_df = tmp_df[tmp_df['Period']=='Comp'].groupby(['Store'],
    ↳as_index=False)['Sum_Gross_Sales'].mean()
tmp_df = pd.DataFrame(tmp_df)
tmp_df.columns = ['Store', 'Avg Gross Sales for Entire Comp Period']

step1_df = pd.merge(sheet_dic['sales'], tmp_df, how='left', on='Store')
del tmp_df

step1_df['Gross Sales Growth'] = step1_df.apply(lambda row:
    ↳round((row['Sum_Gross_Sales']-row['Avg Gross Sales for Entire Comp Period'])/
    ↳row['Avg Gross Sales for Entire Comp Period'], 3), axis=1)
#step1_df['Gross Sales Growth'] = step1_df.apply(lambda row:
    ↳(row['Sum_Gross_Sales']-row['Avg Gross Sales for Entire Comp Period'])/
    ↳row['Avg Gross Sales for Entire Comp Period'], axis=1)
step1_df.head(20)
```

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[3]:
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	Region	Store	Week	Week_Start	Week_End	Period	Sum_Gross_Sales	\
0	Midwest	store_012	49	2014/1/9	2014/1/15	Comp	1682.54	
1	Midwest	store_012	50	2014/1/16	2014/1/22	Comp	2387.10	
2	Midwest	store_012	51	2014/1/23	2014/1/29	Comp	1847.11	
3	Midwest	store_012	52	2014/1/30	2014/2/5	Comp	1192.87	
4	Midwest	store_012	53	2014/2/6	2014/2/12	Comp	2996.67	
5	Midwest	store_012	54	2014/2/13	2014/2/19	Comp	1800.50	
6	Midwest	store_012	55	2014/2/20	2014/2/26	Comp	1087.62	
7	Midwest	store_012	56	2014/2/27	2014/3/5	Comp	1655.80	
8	Midwest	store_012	57	2014/3/6	2014/3/12	Comp	2315.13	

9	Midwest	store_012	58	2014/3/13	2014/3/19	Comp	1002.62
10	Midwest	store_012	59	2014/3/20	2014/3/26	Test	1498.30
11	Midwest	store_012	60	2014/3/27	2014/4/2	Test	2280.74
12	Midwest	store_012	61	2014/4/3	2014/4/9	Test	1639.99
13	Midwest	store_012	62	2014/4/10	2014/4/16	Test	2490.09
14	Midwest	store_012	63	2014/4/17	2014/4/23	Test	1357.89
15	Midwest	store_012	64	2014/4/24	2014/4/30	Test	3552.47
16	Midwest	store_012	65	2014/5/1	2014/5/7	Test	2544.45
17	Midwest	store_012	66	2014/5/8	2014/5/14	Test	1159.21
18	Midwest	store_012	67	2014/5/15	2014/5/21	Test	1241.12
19	Midwest	store_012	68	2014/5/22	2014/5/28	Test	1978.85

	Avg Gross Sales for Entire Comp Period	Gross Sales Growth
0	1796.796	-0.064
1	1796.796	0.329
2	1796.796	0.028
3	1796.796	-0.336
4	1796.796	0.668
5	1796.796	0.002
6	1796.796	-0.395
7	1796.796	-0.078
8	1796.796	0.288
9	1796.796	-0.442
10	1796.796	-0.166
11	1796.796	0.269
12	1796.796	-0.087
13	1796.796	0.386
14	1796.796	-0.244
15	1796.796	0.977
16	1796.796	0.416
17	1796.796	-0.355
18	1796.796	-0.309
19	1796.796	0.101

```
[4]: step1_subdf = step1_df[['Store', 'Avg Gross Sales for Entire Comp Period']].
      ↪drop_duplicates()
      step1_subdf
```

```
[4]:
```

	Store	Avg Gross Sales for Entire Comp Period
0	store_012	1796.796000
20	store_015	1890.905000
40	store_024	1401.447000
60	store_027	1155.485556
77	store_029	2013.836000
...	...	...
1670	store_788	2915.016000
1690	store_796	1884.956000

1710	store_798	2291.301000
1730	store_802	2774.967000
1750	store_808	1803.326000

[89 rows x 2 columns]

```
[5]: tmp_df = step1_df.groupby(['Store', 'Period'], as_index=False)['Gross Sales_Growth'].mean()
step2_df = tmp_df.pivot(index='Store', columns='Period')['Gross Sales Growth'].reset_index()

del tmp_df
step2_df.head(20)
```

```
[5]: Period      Store      Comp      Test
0      store_012 -5.551115e-18  0.098800
1      store_015 -1.110223e-17 -0.542900
2      store_024 -1.000000e-04  0.200100
3      store_027 -1.111111e-04 -0.211625
4      store_029  1.000000e-04 -0.187700
5      store_033 -2.000000e-04 -0.503900
6      store_150  0.000000e+00 -0.153600
7      store_251 -1.110223e-17 -0.032222
8      store_299 -1.000000e-04 -0.080800
9      store_323 -1.387779e-18 -0.115700
10     store_462 -1.000000e-04 -0.319500
11     store_466 -1.000000e-04  0.116600
12     store_478 -2.775558e-18 -0.014000
13     store_479 -4.440892e-17 -0.159900
14     store_484  1.000000e-04 -0.405300
15     store_487 -1.000000e-04 -0.115556
16     store_489  1.000000e-04 -0.213700
17     store_490  2.000000e-04 -0.265500
18     store_491 -1.000000e-04 -0.146500
19     store_509 -2.775558e-18 -0.118333
```

```
[6]: def pair(store, treatment_pairing, control):
    if control == 'Control':
        return f'{store} <-> {treatment_pairing}'
    else:
        return store

tmp_df = sheet_dic['treatment_control_pairs'][['Controls', 'Treatments', 'Region']]
tmp_df.columns = ['Store', 'Treatment-Pairing', 'Region']
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tmp_df = pd.merge(sheet_dic['total_store_list'], tmp_df, how='left',
    ↳on=['Store', 'Region'])
tmp_df['Control-Treatment Pairing'] = tmp_df.apply(lambda row:
    ↳pair(row['Store'], row['Treatment-Pairing'], row['Type']), axis=1)

step3_df = tmp_df[['Store', 'Treatment-Pairing', 'Region', 'Type',
    ↳'Control-Treatment Pairing']]
#step3_df.columns = ['store_no', 'Treatment-Pairing', 'Region', 'Type',
    ↳'Control-Treatment Pairing']

del tmp_df
step3_df.head(20)

```

```

[6]:
      Store Treatment-Pairing  Region  Type Control-Treatment Pairing
0   store_015              NaN  Midwest  Treatment      store_015
1   store_024              NaN  Midwest  Treatment      store_024
2   store_027              NaN  Midwest  Treatment      store_027
3   store_695              NaN  Midwest  Treatment      store_695
4   store_696              NaN  Midwest  Treatment      store_696
5   store_707              NaN  Midwest  Treatment      store_707
6   store_719              NaN  Midwest  Treatment      store_719
7   store_721              NaN  Midwest  Treatment      store_721
8   store_724              NaN  Midwest  Treatment      store_724
9   store_727              NaN  Midwest  Treatment      store_727
10  store_741              NaN  Midwest  Treatment      store_741
11  store_742              NaN  Midwest  Treatment      store_742
12  store_755              NaN  Midwest  Treatment      store_755
13  store_767              NaN  Midwest  Treatment      store_767
14  store_769              NaN  Midwest  Treatment      store_769
15  store_596              NaN    West  Treatment      store_596
16  store_601              NaN    West  Treatment      store_601
17  store_687      store_015  Midwest    Control  store_687 <-> store_015
18  store_728      store_015  Midwest    Control  store_728 <-> store_015
19  store_597      store_024  Midwest    Control  store_597 <-> store_024

```

```

[7]: step4_df = pd.merge(step3_df, step2_df, how='left')
step4_df.columns = ['Store', 'Treatment-Pairing', 'Region', 'Type',
    ↳'Control-Treatment Pairing', 'Avg Growth Comparative Period', 'Avg Growth
    ↳Test Period']

step4_df.to_csv('step4_df.csv')

step4_df

```

```

[7]:
      Store Treatment-Pairing  Region  Type Control-Treatment Pairing \
0   store_015              NaN  Midwest  Treatment      store_015
1   store_024              NaN  Midwest  Treatment      store_024

```

2	store_027	NaN	Midwest	Treatment		store_027
3	store_695	NaN	Midwest	Treatment		store_695
4	store_696	NaN	Midwest	Treatment		store_696
..	...	...	...	...		...
78	store_636	store_769	Midwest	Control	store_636 <->	store_769
79	store_594	store_596	West	Control	store_594 <->	store_596
80	store_526	store_596	West	Control	store_526 <->	store_596
81	store_478	store_601	West	Control	store_478 <->	store_601
82	store_527	store_601	West	Control	store_527 <->	store_601

	Avg Growth Comparative Period	Avg Growth Test Period
0	-1.110223e-17	-0.542900
1	-1.000000e-04	0.200100
2	-1.111111e-04	-0.211625
3	2.000000e-04	-0.406800
4	-9.367507e-18	-0.374500
..	...	...
78	1.000000e-04	-0.262800
79	1.110223e-17	0.042200
80	1.000000e-04	-0.331100
81	-2.775558e-18	-0.014000
82	-1.000000e-04	-0.116700

[83 rows x 7 columns]

```
[8]: step5_df = step4_df.copy()
step5_df['Growth_Difference'] = step5_df.apply(lambda row: row['Avg Growth Test_
->Period'] - row['Avg Growth Comparative Period'], axis=1)
step5_df
```

```
[8]:      Store Treatment-Pairing  Region  Type Control-Treatment Pairing \
0   store_015                NaN  Midwest  Treatment                store_015
1   store_024                NaN  Midwest  Treatment                store_024
2   store_027                NaN  Midwest  Treatment                store_027
3   store_695                NaN  Midwest  Treatment                store_695
4   store_696                NaN  Midwest  Treatment                store_696
..   ...                    ...   ...   ...                    ...
78  store_636            store_769  Midwest  Control  store_636 <-> store_769
79  store_594            store_596    West  Control  store_594 <-> store_596
80  store_526            store_596    West  Control  store_526 <-> store_596
81  store_478            store_601    West  Control  store_478 <-> store_601
82  store_527            store_601    West  Control  store_527 <-> store_601

      Avg Growth Comparative Period  Avg Growth Test Period  Growth_Difference
0                -1.110223e-17                -0.542900                -0.542900
1                -1.000000e-04                 0.200100                 0.200200
2                -1.111111e-04                -0.211625                -0.211514
```

3	2.000000e-04	-0.406800	-0.407000
4	-9.367507e-18	-0.374500	-0.374500
..	...	...	...
78	1.000000e-04	-0.262800	-0.262900
79	1.110223e-17	0.042200	0.042200
80	1.000000e-04	-0.331100	-0.331200
81	-2.775558e-18	-0.014000	-0.014000
82	-1.000000e-04	-0.116700	-0.116600

[83 rows x 8 columns]

```
[9]: from scipy import stats
```

```
[10]: arr_1 = step5_df[step5_df['Type']=='Treatment']['Growth_Difference']
arr_2 = step5_df[step5_df['Type']=='Control']['Growth_Difference']
ttest_results = stats.ttest_ind(arr_1, arr_2, equal_var=False)
ttest_results
```

```
[10]: Ttest_indResult(statistic=-1.1924271050832889, pvalue=0.24692900792352512)
```

```
[11]: f'{round(1 - ttest_results.pvalue, 3)*100}%'
```

```
[11]: '75.3%'
```

```
[12]: tmp_df = sheet_dic['total_store_list']
tmp_df = tmp_df[tmp_df['Type'] == 'Treatment']['Store']
tmp_df.columns = ['Treatments']

tmp_df2 = step5_df[['Store', 'Treatment-Pairing', 'Region',
↳ 'Growth_Difference']]
tmp_df2.columns = ['Controls', 'Treatments', 'Region',
↳ 'Difference_GrossSales_Control']

step6_df = pd.merge(tmp_df, tmp_df2, how='left')

tmp_df.columns = ['Controls']
tmp_df3 = pd.merge(tmp_df, tmp_df2, how='left')
tmp_df3 = tmp_df3[['Controls', 'Difference_GrossSales_Control']]
tmp_df3.columns = ['Treatments', 'Difference_GrossSales_Treatment']

step6_df = pd.merge(step6_df, tmp_df3, how='left')
step6_df = step6_df[['Controls', 'Treatments', 'Region',
↳ 'Difference_GrossSales_Control', 'Difference_GrossSales_Treatment']]
```

```
[13]: step6_df['Lift'] = step6_df.apply(lambda row:
↳ (row['Difference_GrossSales_Treatment']-row['Difference_GrossSales_Control'])/
↳ (1+row['Difference_GrossSales_Control']), axis=1)
```

```
del tmp_df
del tmp_df2
del tmp_df3
```

```
[14]: tmp_df = step1_subdf.copy()
tmp_df.columns = ['Treatments', 'Avg Sales from Comp Period']
step6_df = pd.merge(step6_df, tmp_df, how='left', on='Treatments').
↳drop_duplicates().reset_index()
step6_df['Expected Sales Impact'] = step6_df.apply(lambda row:
↳row['Lift']*row['Avg Sales from Comp Period'], axis=1)

del tmp_df

step6_df
```

```
[14]:
```

	index	Controls	Treatments	Region	Difference_GrossSales_Control \
0	0	store_687	store_015	Midwest	-0.059200
1	1	store_728	store_015	Midwest	0.174100
2	2	store_597	store_024	Midwest	-0.515400
3	3	store_776	store_024	Midwest	-0.130300
4	10	store_597	store_027	Midwest	-0.515400
5	11	store_776	store_027	Midwest	-0.130300
6	18	store_654	store_695	Midwest	-0.351700
7	19	store_550	store_695	Midwest	-0.245300
8	20	store_776	store_696	Midwest	-0.130300
9	22	store_788	store_696	Midwest	0.109300
10	27	store_776	store_707	Midwest	-0.130300
11	30	store_710	store_707	Midwest	0.077900
12	34	store_703	store_719	Midwest	-0.140100
13	35	store_615	store_719	Midwest	-0.254300
14	36	store_733	store_721	Midwest	0.188700
15	37	store_760	store_721	Midwest	-0.166200
16	38	store_751	store_724	Midwest	-0.173200
17	39	store_150	store_724	Midwest	-0.153600
18	40	store_713	store_727	Midwest	-0.222700
19	41	store_700	store_727	Midwest	-0.149700
20	42	store_776	store_741	Midwest	-0.130300
21	47	store_798	store_741	Midwest	0.257300
22	49	store_776	store_742	Midwest	-0.130300
23	54	store_705	store_742	Midwest	0.006300
24	56	store_251	store_755	Midwest	-0.032222
25	57	store_577	store_755	Midwest	0.093900
26	58	store_029	store_767	Midwest	-0.187800
27	59	store_714	store_767	Midwest	-0.275300
28	60	store_033	store_769	Midwest	-0.503700
29	61	store_636	store_769	Midwest	-0.262900

30	62	store_594	store_596	West	0.042200
31	63	store_526	store_596	West	-0.331200
32	64	store_478	store_601	West	-0.014000
33	65	store_527	store_601	West	-0.116600

	Difference_GrossSales_Treatment	Lift	Avg Sales from Comp Period \
0	-0.542900	-0.514137	1890.905000
1	-0.542900	-0.610681	1890.905000
2	0.200200	1.476682	1401.447000
3	0.200200	0.380016	1401.447000
4	-0.211514	0.627086	1155.485556
5	-0.211514	-0.093381	1155.485556
6	-0.407000	-0.085300	1903.757000
7	-0.407000	-0.214257	1903.757000
8	-0.374500	-0.280786	1797.294000
9	-0.374500	-0.436131	1797.294000
10	-0.045500	0.097505	1064.224000
11	-0.045500	-0.114482	1064.224000
12	-0.163800	-0.027561	2381.429000
13	-0.163800	0.121362	2381.429000
14	-0.395500	-0.491461	2323.660000
15	-0.395500	-0.275006	2323.660000
16	0.088000	0.315917	1050.332000
17	0.088000	0.285444	1050.332000
18	-0.083200	0.179467	2003.134000
19	-0.083200	0.078208	2003.134000
20	-0.447700	-0.364953	1590.269000
21	-0.447700	-0.560725	1590.269000
22	-0.413400	-0.325515	2542.136000
23	-0.413400	-0.417072	2542.136000
24	0.038600	0.073180	2178.119000
25	0.038600	-0.050553	2178.119000
26	-0.283200	-0.117459	1450.457000
27	-0.283200	-0.010901	1450.457000
28	0.026100	1.067499	1241.438000
29	0.026100	0.392077	1241.438000
30	-0.241200	-0.271925	1942.662000
31	-0.241200	0.134569	1942.662000
32	-0.299600	-0.289655	2044.152000
33	-0.299600	-0.207154	2044.152000

	Expected Sales Impact
0	-972.184044
1	-1154.738851
2	2069.491278
3	532.572420
4	724.589377

5	-107.900972
6	-162.390501
7	-407.893874
8	-504.655852
9	-783.855438
10	103.767041
11	-121.834346
12	-65.635385
13	289.016125
14	-1141.988872
15	-639.020434
16	331.817511
17	299.811214
18	359.497225
19	156.660486
20	-580.374130
21	-891.704164
22	-827.502244
23	-1060.254873
24	159.395299
25	-110.110596
26	-170.368872
27	-15.811522
28	1325.234440
29	486.739360
30	-528.257926
31	261.422817
32	-592.099200
33	-423.454625

```
[15]: avg_lift = round(step6_df['Lift'].mean(), 3)*100
      f'{avg_lift}%'
```

```
[15]: '-1.6%'
```

```
[16]: avg_expected_sales_impact = round(step6_df['Expected Sales Impact'].mean(), 1)
      avg_expected_sales_impact
```

```
[16]: -122.4
```

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[ ]:
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